

Equation (36) is a special form, for the zero structural damping case $\mathbf{d} = 0$, of the following:

$$\mathbf{a}_m \ddot{\mathbf{q}} + (\rho V \mathbf{b} + \mathbf{d}) \dot{\mathbf{q}} + (\rho V^2 \mathbf{c} + \mathbf{e}) \mathbf{q} = \mathbf{0} \quad (37)$$

where $\mathbf{a}_m$ is the mass, or inertia, matrix, $\mathbf{b}$ is the aerodynamic damping matrix, $\mathbf{c}$ is the aerodynamic stiffness matrix, $\mathbf{d}$ is the structural damping and $\mathbf{e}$ is the stiffness matrix. Now, considering the availability of modal data and assuming that the measured damping is dependent solely on the structure itself, $\mathbf{d}$ can be built from the uncoupled modal damping assumption in [52]:

$$\mathbf{d} = \boldsymbol{\phi}^{-T} \mathbf{d}_n \boldsymbol{\phi}^{-1} \quad \text{for } \mathbf{d}_n = 2\zeta_n \omega_n \mathbf{a}_n \quad (38)$$

where the subscript n identifies the uncoupled matrix, ω_n the natural frequency, ζ_n the damping ratio and $\boldsymbol{\phi}$ the mode shape. Hence, Equations (36) and (38) can be combined to assemble Equation (37). Given full knowledge of the wing geometric characteristics, three properties remain to be defined: M_θ , EI , and GJ . M_θ is the unsteady aerodynamics term and it is defined from oscillatory aerodynamics [49]:

$$M_\theta = 2\pi \left[-\frac{k}{2} \left(\frac{1}{2} - a \right) + kF \left(a + \frac{1}{2} \right) \left(\frac{1}{2} - a \right) + \frac{G}{k} \left(\frac{1}{2} + a \right) \right] \quad (39)$$

where f_k is the reduced frequency, a is the ratio between c and the flexural axis position, and F and G are, respectively, the real and imaginary part of Theodorsen's function, $C(f_k)$, such that:

$$C(f_k) = F(f_k) + jG(f_k) = \frac{H_1^{(2)}(f_k)}{H_1^{(2)}(f_k) + jH_1^{(2)}(f_k)} \quad (40)$$

where $H_n^{(2)}(f_k)$ are Hankel functions of the second kind and j is the imaginary number.

Concerning the bending and torsional stiffness, EI and GJ , let us consider the still air case where $\mathbf{b}$ and $\mathbf{c}$ are zero. Equation (37) then becomes a simple mass-spring-damper system:

$$\mathbf{a}_m \ddot{\mathbf{q}} + \mathbf{d} \dot{\mathbf{q}} + \mathbf{e} \mathbf{q} = \mathbf{0} \quad (41)$$

The natural frequencies can then be easily extracted through eigenanalysis. Hence, by having a set of experimental ω_n it is possible to define the EI and GJ of the equivalent system by minimising its squared difference to the experimental ω_n . Thus, by using this and the identified ζ_n for Equation (38), an aeroelastic model can be defined starting from experimental data. To study the system stability, the eigenanalysis of Equation (41) can be solved iteratively with the well-known p - k method [53] to find the divergence and flutter onset speeds. The p - k method is based on the hypothesis that pure harmonic aerodynamics can be used as a good approximation for lightly damped harmonic motions. This allows the computation of the aerodynamic transfer matrix at a complex frequency $p = \delta \pm jk$, such that $p \approx jk$. In simple terms, the real part, i.e. the damping, is neglected. A widely accepted workflow of the p - k method [49] can be summarised as follow:

1. Initiate an estimation, usually the still air value, of p , said $p_0 = \delta \pm jk_0$
2. Evaluate the aerodynamics, in our case M_θ
3. Solve the eigenvalue (λ) problem for Equation (37) and obtain a new set of λ , $p_1 = \delta \pm jk_1$
4. Iterate between 2 and 3 until $k_n \approx k_{n-1}$

From the λ obtained after convergence, it is possible to build ω_n , ζ_n , $\text{real}(\lambda)$ and $\text{imag}(\lambda)$ vs air-speed (U_∞) plots, which can be used to graphically portray divergence or flutter speed, whichever is detected first. Particularly, critical speeds are identified for ζ_n approaching zero or for $\text{real}(\lambda)$ zero crossings, since both cases are interpreted as instability in the system. Particularly, for flutter, only the $\text{real}(\lambda)$ zero crossing condition needs to be satisfied, while for divergence, $\text{imag}(\lambda)$ must be zero.